

Simple DUAL-WAY AMPLIFIER For Microphone And Guitars

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A self-made audio power amplifier for a guitar or violin should be simple, low-cost and reliable. It is good to have two separate loudspeakers pointing in different directions. It is better when the amplifier has two separate channels working independently, either paralleled or as a bridged amplifier. This article presents a simple two-channel outputs amplifier using two LM1875 audio amplifiers.

Circuit and working

Circuit diagram of the simple dual-way audio power amplifier for a microphone and two guitars built around three transistors and two LM1875 ICs is shown Fig. 1.

The amplifier has two different

input stages that use transistor T1 for a microphone and T2 for two guitar inputs. The mixing stage is built around T3 and two power amplifiers around IC1 and IC2. You can use any of the power amplifier ICs listed in Table 1.

There are many low-cost power amplifier ICs having 10W - 50W

or more peak-output power. Even though some of these ICs produce sufficient power for small guitars and violins, it is a good idea to combine two ICs for more output power and the possibility to drive one or two systems with loudspeakers.

When a single load is driven, both the amplifiers could be bridged.

TABLE 1
**MAIN PARAMETERS OF AUDIO POWER AMPLIFIERS
RELEVANT TO THE PROJECT**

IC	Max power supply $\pm V_{ee}$	Iout Peak	Pd	Tjmax	Pout(1yp)/RL/ $\pm V_{ee}$
LM1875	$\pm 30V$ (60V)	3A	20W	+150°C	20W/8-ohm/ $\pm 25V$
TDA2030	$\pm 18V$ (36V)	3.5A	20W	+150°C	14W/4-ohm/ $\pm 14V$
TDA2040	$\pm 20V$ (40V)	4A	25W	+150°C	30W/4-ohm/ $\pm 17V$
TDA2050	$\pm 25V$ (50V)	5A	25W	+150°C	35W/4-ohm/ $\pm 18V$

Note. Pd is power dissipation, Tjmax is maximal junction temperature, Pout is output power and RL is load resistance for the IC

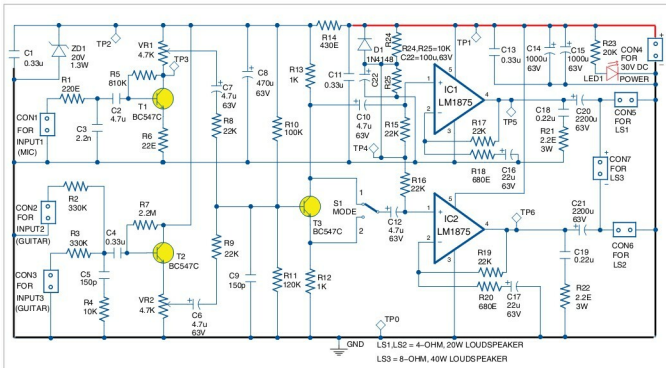


Fig. 1: Circuit diagram of the simple dual-way amplifier for a microphone and two guitars